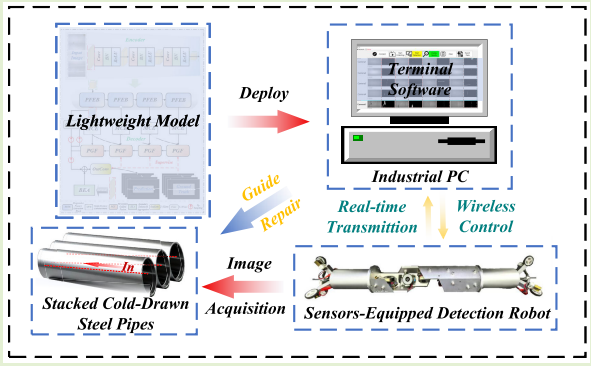
A Lightweight Perception Enhancement Network for Real-Time and Accurate Internal Surface Defect Detection of Cold-Drawn Steel Pipes

You Tan, Kechen Song, *Senior Member, IEEE*, Hongshu Chen, Yu Zhang, and Yunhui Yan

***Abstract*—The detection of internal surface defects in cold-drawn pipes is challenging. In recent years, as the production demands for cold-drawn steel pipes have steadily grown, there has been an urgent need for an efficient detection approach that balances accuracy and real-time performance in industrial environments. Although several existing deep learning-based methods have achieved high accuracy in surface defect detection, they often need substantial computational costs to extract rich feature representations, which inevitably slows down the inference process and leads to low detection efficiency. Moreover, internal defects of cold-drawn pipes typically exhibit challenges, which may further degrade the performance of existing models. To address these challenges, we propose a lightweight perception enhancement network (LPENet) to effectively balance efficiency and accuracy. Specifically, we introduce a progressive feature extraction (PFE) backbone that enhances contextual perception from local to global scales. Furthermore, we design a multiscale context enhancement (MCE) module to enrich the feature representation and a boundary-enhanced aggregation (BEA) module to strengthen fine-grained feature awareness. In addition, we propose a perception-guided fusion (PGF) strategy to facilitate interaction between shallow and deep features. We deploy LPENet in combination with a pipe internal surface detection (PISD) robot, achieving wireless and efficient defect detection in real-world steel pipe factories. In extensive experiments on the SSP2000 dataset, LPENet achieves the best balance between detection accuracy and efficiency. The source code is publicly available at https://github.com/VDT-2048/LPENet.**



***Index Terms*— Cold-drawn steel pipes, detection efficiency, internal surface defect detection, lightweight salient object detection (SOD), vision sensors.**

I. Introduction

COLD-DRAWN steel pipes, as the critical industrial material for high-pressure fluid transmission systems, exhibit exceptional compressive strength and corrosion resistance [1]. To ensure the safety and stability of the pipes during subsequent service, surface defect detection is indispensable. Compared with external defects, internal surface defects are much more difficult to detect due to the enclosed internal environment. If not repaired in time, such defects may lead to severe seriously economic losses or even safety accidents. Therefore, detecting internal surface defects in cold-drawn steel pipes is both critically important and challenging.

In recent years, the sensor technology [2] has significantly improved data acquisition quality, while machine vision-based detection methods have gained increasing attention in both detection performance and the visualization of detection results. Driven by the rapid progress of deep learning, numerous pixel-level surface defect detection approaches have been proposed, among which salient object detection (SOD) methods have demonstrated high-precision detection performance. High-accuracy detection typically focuses on enhancing the spatial localization accuracy and contour representation of defect regions, including, but not limited to, backbone architectures based on convolutional neural networks and Transformer, which can, respectively, provide effective multiscale feature extraction and long-range dependence modeling capabilities. Building upon these techniques, many studies have been conducted. For example, Song et al. [3]

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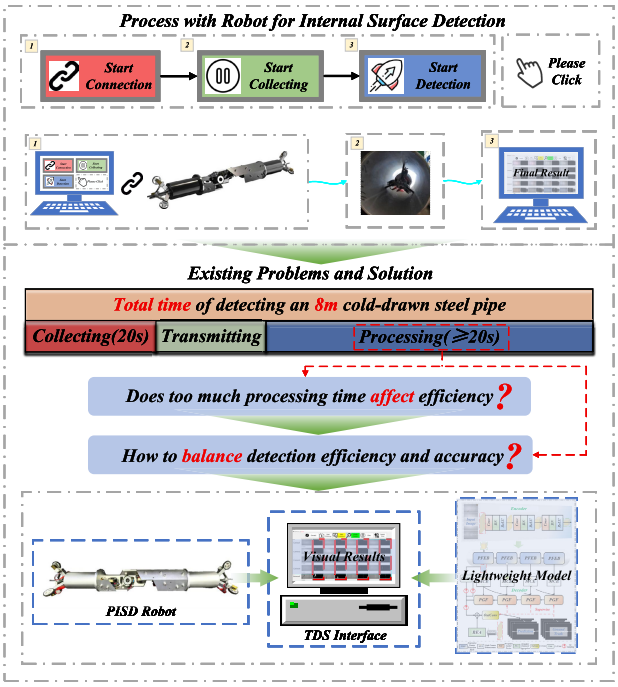


Fig. 1. Process with robot for internal surface detection and existing problems and solution.

treated strip steel surface defects as salient objects and proposed encoder–decoder residual network (EDRNet), which can accurately segment complete defect targets. Similarly, Cui et al. [4] proposed autocorrelation-aware aggregation network (A3Net), which incorporates a scale-interaction module to aggregate features of defects at different scales on the strip steel surface. To promote the construction of smart factories and realize automatic detection process [5], [6], the integration of pipe robots equipped with vision sensors, leveraging deep learning models has gradually attracted increasing attention and is considered to have substantial application potential, as illustrated in Fig. 1. For example, Chen et al. [7] proposed self-reinforcing perception coordination network (SRPCNet), by image acquisition capability of pipe robots and transmit the collected images to terminal units for processing, effectively addressing the challenge of hybrid camouflage defects on steel pipe internal surface. However, these defect detection methods demonstrate high accuracy; they often require considerable computational costs to learn rich feature representations, inevitably prolonging the processing time and reducing inference speed, which, in turn, lowers detection efficiency. As the production demands for cold-drawn steel pipes steadily grow, the requirements for production efficiency have become increasingly stringent. Therefore, an effective defect detection model should take both accuracy and real-time performance into account. In other words, a well-balanced tradeoff between efficiency and precision is essential to support practical applications.

In contrast, lightweight detection methods are playing an increasingly important role, concerning model compactness and computational efficiency, which makes them more favorable for practical deployment and real-time detection. In recent years, extensive research has been devoted to lightweight detection approaches aiming to reduce model parameters and computational cost through techniques, such as depthwise separable convolution (DSConv) [8] and model pruning [9]. Shen et al. [10], inspired by the principle of DSConv, proposed a multiscale interactive network called MINet, which effectively represents strip steel defect regions with a reduced model size. Li et al. [11] developed a lightweight SOD method that jointly employs a scale association module and a dense lightweight attention module to achieve efficient detection. However, the above surface defect detection methods are mainly designed for planar structures such as strip steel surfaces, where they can deliver promising performance. This work focuses on the detection of defects on the internal surface of steel pipes, where the following two issues must be carefully addressed.

1) Compared with planar structures such as strip steel, cold-drawn steel pipes present a fully enclosed internal space, where image acquisition is inevitably constrained by factors such as narrow spatial conditions, extreme light, and limited viewing angles—making the detection process inherently challenging.
2) The aforementioned detection methods show certain advantages when applied to strip steel surface defects. However, compared with strip steel defects, internal surface defects in steel pipes typically exhibit lower contrast, more reflective, and in some cases, small-sized object defects. These challenges can lead to performance degradation during model processing.

Therefore, the development of a real-time internal surface defect detection system for steel pipes, which achieves a balance between detection accuracy and efficiency, is of great significance. To further align with industrial scenarios, and considering the restriction of a 100–150-mm detection inner diameter range, we designed a pipe internal surface detection (PISD) robot system and a terminal detection software (TDS), as illustrated in Fig. 1. The PISD robot system is equipped with five sets of vision-integrated detection units, each consisting of a CMOS sensor, an embedded board, a ring light source, and a helical lens, as illustrated in Fig. 2. Unlike previous works [12], [13], [14], operators only need to issue commands through the TDS to wirelessly control the PISD robot, enabling it to move flexibly inside the pipe and capture images of all regions of the internal surface. The acquired images are, then, compressed and wirelessly transmitted to an industrial PC, where a deep neural network ensures accurate and efficient defect detection. To further optimize model size and inference speed, making the detection method more suitable for real industrial scenarios, we propose the first lightweight network, called lightweight perception enhancement network (LPENet), for internal surface defect detection of cold-drawn steel pipes. LPENet maintains a compact model size while achieving a balanced tradeoff between detection accuracy and efficiency. Specifically, we first design a two-stage encoder based on the progressive feature extraction (PFE) module. To ensure sufficient feature perception capability, we propose the multiscale context enhancement (MCE) module. Next, we adopt the perception-guided fusion (PGF) module strategy to fully exploit complementary information across different feature levels, thereby enhancing the integrity of detection results.

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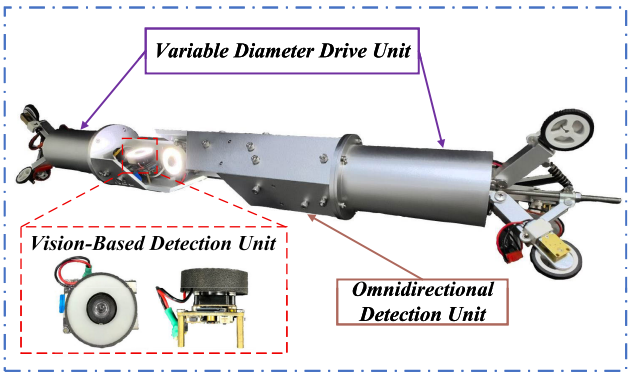


Fig. 2. Description of the internal surface defect detection robot.

Finally, the boundary-enhanced aggregation (BEA) module is incorporated to strengthen the network's edge perception for defect regions.

In summary, the contributions of this article can be summarized as follows.

1) To the best of authors' knowledge, we propose the first lightweight SOD network for detecting internal surface defects of cold-drawn steel pipes. Compared with 17 state-of-the-art SOD methods, LPENet achieves the best balance between efficiency and accuracy.
2) By leveraging the wireless control and real-time transmission capabilities of the TDS, we deploy the proposed PISD robot in combination with the LPENet and test against existing industrial detection methods, achieving a significant improvement in detection efficiency.
3) We propose a PFE-block-based backbone that implements a local-to-global feature extraction strategy, while maintaining a low parameter count.
4) We design three novel modules: the MCE module and the BEA module, which are designed to capture multiscale features and enhance fine-grained boundary feature perception; and the PGF module facilitates improving the completeness of the detection results.

II. Related Work

A. SOD Methods

SOD aims to simulate the human visual perception system to capture the most important regions in a given image. In recent years, many researchers have focused on natural scene objects as the primary targets of interest. For example, Zhang et al. [15] proposed FasterSal, an RGB-D SOD framework that mitigates inconsistencies between RGB-D and RGB pretrained encoders, effectively localizing salient objects. Zhong et al. [16] introduced a cross-modal integration module in lightweight and efficient network for salient object detection (LESOD), achieving efficient and accurate detection. Jin et al. [17] developed local and global perception network (LGPNet), an RGB-T SOD network that performs multimodal feature fusion to enhance robustness in complex environments. Beyond natural scenarios, SOD has promising applications in industrial scenarios. Niu et al. [18] constructed the RSDDS-113 steel rail dataset using a binocular line-scanning system and proposed an unsupervised stereo saliency detection method. Zhou et al. [19] presented extraction-amplification-fusion network (EAFNet), addressing the challenges of insufficient feature information and interference in tiny cracks detection by employing feature extraction, amplification, and fusion modules to improve detection performance. Yang et al. [20] integrated a blade-in-service dataset and proposed an improved edge-guided channel enhancement network based on a Transformer architecture, successfully localizing and segmenting blade defects.

B. Lightweight Detection Methods

To maintain detection accuracy while achieving low computational cost, many researchers have focused on designing lightweight detection methods, which have gradually become an important choice for deep learning applications in resource-constrained scenarios. In the field of remote sensing, Han et al. [21] proposed a lightweight region awareness and multipaths enhancement network (RAMENet), which achieves multigranularity feature aggregation through cross-layer feature interaction, significantly enhancing fine-detail representation while preserving object structure integrity. Li et al. [22] introduced speed-oriented lightweight network (SOLNet), a speed-oriented method that performs lightweight and efficient decoding through simple cascade aggregation operations. Shen et al. [23] proposed a lightweight saliency model based on a fully compressed multiscale module, enabling real-time and accurate segmentation of salient objects from optical remote sensing images. In the field of defect detection, Pang et al. [24] developed a YOLOv9-DC-based tracker, meeting the efficiency requirements for real-time tracking of multiple crack defects. Mo et al. [25] proposed a SE-ENv2 GC-Neck TSCODE-YOLO (SGT-YOLO) framework based on YOLOv5s for microdefects on printed circuit board (PCB) boards, introducing a task-specific context decoupling head to effectively extract both positional and class information of defects. Recently, significant progress has also been made in lightweight camouflaged object detection. Ren et al. [26] proposed efficient skeleton-guided network (ESNet), which enhances the integrity of camouflaged object detection while maintaining sensitivity to internal object structures. Li et al. [27] improved the backbone features at each stage using information from adjacent stages, achieving more comprehensive feature representations with lightweight parameters.

C. Surface Defect Detection

Surface defect detection plays a crucial role in ensuring product quality and production safety. Compared with traditional methods, deep learning-based approaches combined with machine vision not only guarantee high-quality data acquisition, but also enable intelligent detection capabilities. Zhou et al. [28] reviewed common deep learning methods for ceramic surface defect detection and summarized their research progress and application performance. Gong et al. [29] simplified the YOLO algorithm to extract features from multiple photovoltaic panel surface defects, achieving high accuracy while reducing computational resource requirements. Liu et al. [30] proposed a dual-stream enhanced object detection strategy, which, unlike YOLO-based methods [31], [32], can accurately and in real-time identify steel strip

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surface defects with fewer parameters. Zhang et al. [33] introduced a dual feature-integration network based on ResNet18, enabling fast and precise small-sample classification of steel strip surface defects. Guo et al. [34] proposed a GAN-based welding defect detection method with transfer learning, effectively addressing the imbalance of weld radiography data and improving detection accuracy and robustness. Song et al. [35] developed a cross-granularity few-shot semantic segmentation paradigm, tackling the generalization issues caused by scarce internal surface defect samples in cold-drawn steel pipes under few-shot or supervised learning settings, achieving promising detection results.

III. Method

For SOD, the key lies in accurately locating and segmenting visually distinctive regions in a given image. This requires the model not only to effectively distinguish salient targets from the background, but also to possess sufficient feature perception capability. In this section, we give the original input feature $F_o \in \mathbb{R}^{C\times H\times W}$, and first take a brief introduction for our model in Section III-A. In Section III-B, we present the PFE backbone. In Section III-C, we present the MCE module. In Sections III-D and III-E, we detail the PGF and BEA modules. In Section III-F, we introduce the loss function of our network.

A. Architecture Overview

As illustrated in Fig. 3, the LPENet is proposed. To achieve both high-speed and high-accuracy detection for internal surface defect detection of cold-drawn steel pipes, lightweight deployment is essential. In the feature extraction stage, the PFE block is first introduced, and based on this, the PFE backbone is designed to obtain the basic feature representations $F^{(i)}$ ($i = 1, 2, 3,$ and 4). Following feature extraction, the MCE module is developed to effectively capture multiscale contextual information, thereby improving the accurate perception of salient objects. Subsequently, the PGF module is introduced to focus on the correlation among features at different levels, enabling cross-layer guidance and the fusion of shallow details with deep semantic information. To further enhance the representation of edge details, the BEA module is proposed, which aggregates contextual and boundary information through a multistep enhancement mechanism, enabling fine-grained defect detection. Moreover, each intermediate layer's output features are explicitly supervised, and the final saliency prediction map is generated. It is worth noting that the parameters of all the aforementioned modules are mutually independent, with no weight sharing.

B. PFE Backbone

Balancing detection accuracy and speed remains a critical challenge for achieving efficient real-time inspection, and this is inherently dependent on the rational design of the feature extraction backbone. Existing surface defect detection methods are often built upon backbones such as VGG16 [36] and ResNet50 [37]. While these architectures provide effective multiscale feature extraction capabilities, they inevitably involve a large number of parameters and floating-point operations, which substantially increase the network's

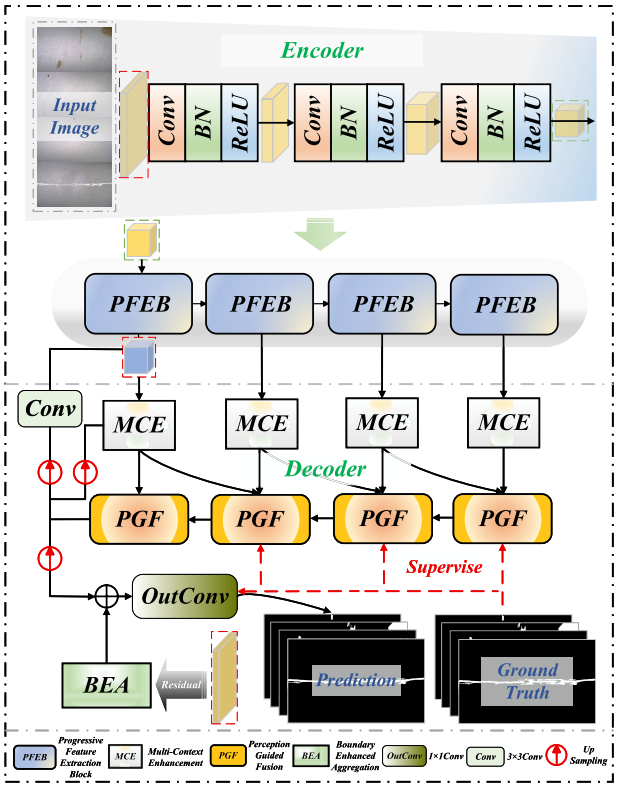


Fig. 3. Overall framework of the proposed LPENet.

processing time. Inspired by prior works [38], [39], we explore several efficient strategies proposed in recent years [40], [41] and introduce a PFE block designed for deployment in a real-time, lightweight backbone network dedicated to internal surface defect detection of steel pipes, as illustrated in Fig. 4.

We propose a PFE backbone that maintains a low parameter count while following a local-to-global feature extraction paradigm. The PFE backbone can be conceptually divided into two stages: the first stage reduces the spatial resolution through downsampling while expanding the channel dimension of features, and the second stage focuses on extracting enriched feature representations. To minimize parameters and computational complexity while retaining the effective feature extraction properties of the classic VGG16, we introduce several modifications. In the first stage, a sequence of downsampling operations is employed to preliminarily capture local textures and edge structures. In the second stage, the PFE block is applied to enhance global contextual perception. As shown in Fig. 4, the PFE module primarily consists of an initial normalization layer, a fully connected layer, a fusion layer, and a final output layer. The backbone operates as follows: first, three conventional downsampling layers are applied to extract local textures and edge features, yielding the initial feature map F_c. The extracted features are then processed, while preserving their spatial positions through the PFE block that performs nonlinear mapping on the channel features at each location. Specifically, a reshape operation is first applied to transform the features along the channel dimension, followed by a normalization step. Subsequently, a classical fully connected operation is employed to provide higher dimensional feature representations. The resulting features are,

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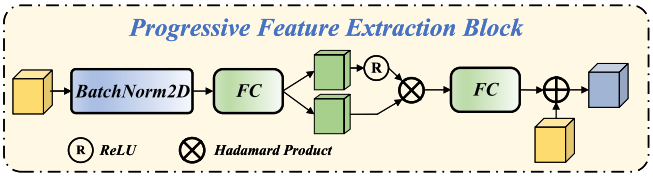


Fig. 4. Structure of the PFE block.

then, evenly split into two parts along the channel dimension, and a gating-unit-like operation is applied to achieve feature fusion while effectively suppressing redundant information. Finally, residual connections with the input features are incorporated to facilitate the acquisition of contextual information. Following the approach in prior studies [38], four PFE blocks are sequentially arranged in this stage to progressively capture contextual information $F^{(i)}$ ($i = 1, 2, 3,$ and 4) for subsequent decoding. It is worth noting that the linear arrangement of the four PFE blocks in this stage contributes to low latency [10]. Moreover, this lightweight design combines staged processing with local and global feature fusions, allowing the network to retain a low parameter count while still providing sufficiently expressive feature representations.

C. Multiscale Context Enhancement

The internal surface defects of cold-drawn steel pipes exhibit significant scale variability, with some targets being extremely small, which undoubtedly increases the detection challenge. Therefore, it is essential to ensure that the detection method maintains strong feature perception capabilities and sufficient attention to local detail features. According to previous studies [42], [43], dilated convolution (DConv) can effectively enlarge the receptive field while preserving the continuity of feature map information without increasing computational cost. In addition, compared with traditional convolution operations, DSConv can significantly reduce the number of parameters while enhancing the expressive capacity of features. Notably, the advantage of both lies in their ability to extract rich multiscale contextual information and local detail features while maintaining a low model parameter count. Inspired by this, and to achieve stronger multiscale feature capture capability, we propose a lightweight MCE module, as illustrated in Fig. 5. Given an input feature map $F^{(i)}$ ($i = 1, 2, 3,$ and 4), the feature map is evenly divided along three dimensions into three parts, each processed by a distinct operation: a 1×1 convolution operation, a cross-scale refinement, and a cross-scale perception. The detailed process is as follows:

$$F_{\text{mce}}^{(i)} = F_{\text{conv}}^{(i)} + F_{\text{csr}}^{(i)} + F_{\text{csp}}^{(i)} \tag{1}$$

where $F_{\text{mce}}^{(i)}$ denotes the features produced by the MCE. $F_{\text{csr}}^{(i)}$ and $F_{\text{csp}}^{(i)}$ are the features produced by the cross-stage refinement (CSR) and cross-scale perception (CSP) strategy, and $F_{\text{conv}}^{(i)}$ is the features produced by a 1×1 convolution operation.

1) Cross-Scale Refinement: By reasonably using a combination of DConv operations with different dilation rates, we enhance the receptive field of the model while ensuring the continuity of feature information. Inspired by methods

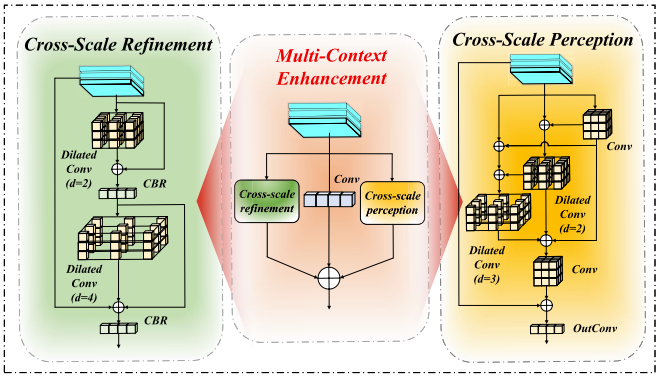


Fig. 5. Structure of the MCE module.

such as DenseNet [44], we propose a CSR module; the CSR is designed for object refinement. In the CSR, two DSConv3×3 layers with dilation rates of 2 and 4 are employed to construct a dense refinement layer. These are followed by two common convolution-batch normalization-relu operation (CBR) layers to integrate the captured features. The process can be formulated as follows:

$$F_{\text{csr}}^{(l-1)} = \text{CBR}\left[F_i + D_r(F_i)\right] \tag{2}$$

$$F_{\text{csr}}^{(i)} = F_{\text{csr}}^{(l)} = \text{CBR}\left[F_{\text{csr}}^{(l-1)} + D_r\left(F_{\text{csr}}^{(l-1)}\right) + F_i\right] \tag{3}$$

where $F_{\text{csr}}^{(i)}$ is the features produced by CSR strategy, and CBR[$*$] represents the performing convolution, batch normalization, and relu operation. $D_r(*)$ denotes the DConv, and r is the dilated rate.

2) Cross-Scale Perception: To further enlarge the receptive field and generate more effective contextual information, three dilated DSConv3×3 branches are applied to the input features to extract different contextual representations, where the dilation rates of the DSConv layers in each branch are set to 1–3, respectively. Unlike the approaches in [45] and [46], the three branches are organized in a cascaded manner, where the input of branch $i+1$ also receives the output of branch i and $i-1$. This design progressively expands the receptive field, thereby enhancing multiscale perception capability. The process can be formulated as follows:

$$F_{\text{csp}}^{(h)} = \begin{cases} \mathbb{D}_1\left[F_i\right], & h = 1 \\ \mathbb{D}_2\left[F_i + F_{\text{csp}}^{(h-1)}\right], & h = 2 \\ \mathbb{D}_3\left[F_i + F_{\text{csp}}^{(h-1)} + F_{\text{csp}}^{(h-2)}\right], & h = 3 \end{cases} \tag{4}$$

$$F_{\text{csp}}^{(i)} = \text{OutConv}\left[D_1\left[F_i + \left(\Sigma_{h=1,2,3}\left(F_{\text{csp}}^{(h)}\right)\right)\right]\right] \tag{5}$$

where $F_{\text{csp}}^{(i)}$ is the features produced by CSP strategy, $D_r[*]$ and r also denote the DConv and the dilated rate, respectively, and OutConv represents a 1×1 convolution operation.

D. Perception-Guided Fusion

There are a large number of continuous and consecutively distributed defect regions on the internal surface of the cold-drawn steel pipes, such as corrosion and extensive oxide scale, which substantially increase the difficulty of achieving complete defect detection. To address this challenge, we propose the PGF module, as illustrated in Fig. 6, to maximize

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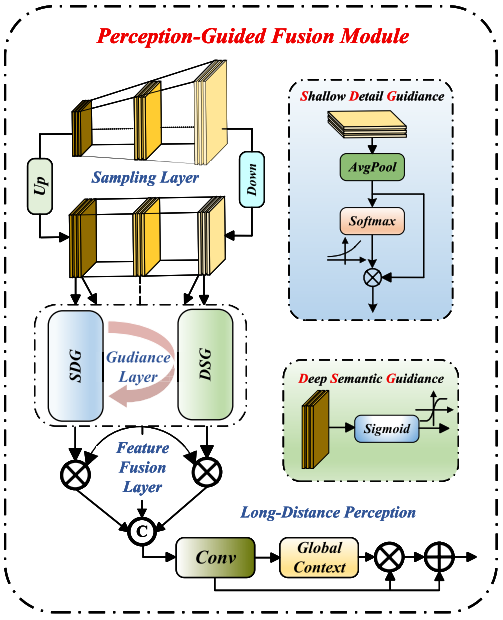


Fig. 6. Structure of the PGF module.

the global perception of extensive defect areas. Globally, the PGF module consists of a sampling layer, a guidance layer, a feature fusion layer, and a long-distance perception layer.

From a technical standpoint, we first apply tensor alignment to the features enhanced by the MCE module. Considering that multiscale information from adjacent feature layers should maintain strong correlations, we perform upsampling and downsampling operations on the subsequent and preceding feature layers, respectively. Then we employ the shallow detail guidance (SDG) and deep semantic guidance (DSG) strategies to facilitate the interaction between shallow details and deep semantic information across adjacent features. Specifically, in the PGF module, the fused features from deeper semantic layers are used to guide the input for shallow detail fusion, while the fused features from shallow detail layers guide the input for deep semantic fusion, thereby achieving comprehensive multilevel feature-guided fusion. Inspired by the work in [47] and [48], to further strengthen the perception of long-range dependencies, we integrate a global context module after feature fusion to establish richer global contextual information. The global process can be formulated as follows:

$$F_{\text{pgf}}{}^{(i)} = F^{*}_{\text{pgf}}{}^{(i)} + \text{Conv}\left(F^{(i)}_{\text{fusion}}\right) \tag{6}$$

$$F^{*}_{\text{pgf}}{}^{(i)} = \text{GC}\left[\text{Conv}\left(F^{(i)}_{\text{fusion}}\right)\right] \odot \text{Conv}\left(F^{(i)}_{\text{fusion}}\right) \tag{7}$$

where $F_{\text{pgf}}{}^{(i)}$ is the features produced by PGF module, GC$(*)$ represents the global context operation, Conv$(*)$ means 1×1 convolution, $\odot$ operation is the Hadamard product, and $F^{(i)}_{\text{fusion}}$ is produced by feature fusion layer

$$F^{(i)}_{\text{fusion}} = \begin{cases} \mathbb{P}_i\left[F^{*}_{\text{mce}}{}^{(i)}, F_{\text{pgf}}{}^{(i+1)}\right], & i = 1 \\ \mathbb{P}_i\left[F^{*}_{\text{mce}}{}^{(i)}, F_{\text{pgf}}{}^{(i+1)}, F^{*}_{\text{mce}}{}^{(i-1)}\right], & i = 2, 3 \\ \mathbb{P}_i\left[F^{*}_{\text{mce}}{}^{(i)}, F^{*}_{\text{mce}}{}^{(i-1)}\right], & i = 4 \end{cases} \tag{8}$$

where $\mathbb{P}_i[*]$ means the specific computational process performed in the PGF module, and $F^{*}_{\text{mce}}{}^{(i)}$ denotes the multilevel MCE features by tensor alignment

$$\mathbb{P}_i\left[F_m, F_s, F_d\right] = \text{Conv}[\mathbb{C}(F_m, \text{DSG}\left(F_d\right) \odot F_s, \text{SDG}\left(F_s\right) \odot F_d)] \tag{9}$$

$$\mathbb{P}_i\left[F_s, F_d\right] = \text{Conv}\left[\mathbb{C}\left(\text{DSG}\left(F_d\right) \odot F_s, \text{SDG}\left(F_s\right) \odot F_d\right)\right] \tag{10}$$

where F_m, F_s, and F_d represent the primary feature at the ith layer, the shallow feature, and the deep feature from the adjacent layer, respectively. $\mathbb{C}$ means the classic concatenate operation. DSG$(*)$ and SDG$(*)$ are the deep semantic guidance and shallow detailed guidance, respectively. Furthermore, the guidance layer operation can be formulated as follows:

$$\text{SDG}\left(F_s\right) = \text{Sig}\left(F_s\right) \tag{11}$$

$$\text{DSG}\left(F_d\right) = \text{Sfx}\left(\text{Avgpool}\left(F_d\right)\right) \odot \text{Avgpool}\left(F_d\right), \tag{12}$$

where Sig$[*]$ represents the classic Sigmoid operation, and Sfx$[*]$ denotes the softmax operation.

E. Boundary-Enhanced Aggregation

Due to the inherently high surface smoothness and metallic reflectivity of cold-drawn steel pipes, specular reflections are inevitable. Such reflections can reduce the contrast between defects and the background in certain images. When a defect is located within a reflective region, both the background and defect textures become attenuated, which may prevent complex fine-grained features and blurred boundaries from being adequately captured, thereby diminishing the defect's saliency. Upsampling the output of the mask decoder to the input resolution to extract sufficient fine-grained details, based on prior research experience, can incur excessive computational overhead and degrade inference speed. To address this issue, we propose a BEA module to improve fine-grained feature perception, as illustrated in Fig. 7. In this module, we apply a 3×3 average pooling followed by a 1×1 convolution at each layer to extract detailed information. On this basis, and to further enhance the edge perception capability for defect regions, we introduce a boundary-enhancement block (BE-Block). Qualitatively, the BE-Block can strengthen boundary information within the feature maps, enabling the network to more clearly distinguish targets from the background in subsequent processing stages. It can be formulated as follows:

$$F_{b_1} = F_b - \text{Avgpool}\left(F_b\right) \tag{13}$$

$$F_{\text{be}} = F_b \odot \text{Conv}\left(F_{b_1}\right) + F_b. \tag{14}$$

Unlike the previous works [4], [7], $F_{\text{pgf}}{}^{(i)}$ is not, then, used as the input to the BEA module. We use the original feature maps F_o to the BEA module. We first perform a convolution operation to obtain local features. The process can be formulated as follows:

$$F_{\text{bla}} = \text{InConv}\left(F_o\right) \tag{15}$$

$$F^{(i)}_{\text{bl}} = \begin{cases} \text{ApConv}\left(F_{\text{bla}}\right), & i = 1 \\ \text{ApConv}\left(F^{(i)}_{\text{bl}}\right), & i = 2, 3 \end{cases} \tag{16}$$

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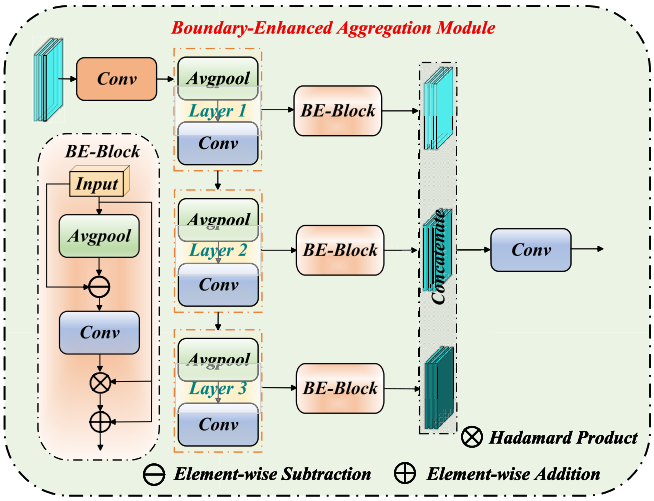


Fig. 7. Structure of the BEA module.

$$F_{\text{bea}}^{(i)} = \text{BE}\left(F_{\text{bl}}^{(i)}\right) \tag{17}$$

$$F_{\text{bea}}^{(i)} = \text{Conv}\left[\mathbb{C}\left(F_{\text{bea}}^{(1)}, F_{\text{bea}}^{(2)}, F_{\text{bea}}^{(3)}\right)\right] \tag{18}$$

where InConv[$*$] represents the 3×3 convulution, ApConv[$*$] represents performing Avgpool and 1×1 convolution, and BE[$*$] denotes the operation by BE-Block.

F. Supervision Strategy

For the internal surface defect detection task, we must maintain consistency with the ground-truth maps during both training and inference; the feature channels and spatial dimensions are adjusted accordingly. Therefore, supervision is provided using a combination of binary crossentropy loss (BCE) loss and intersection over union (IOU) loss, computed as follows:

$$\text{Loss} = L_{\text{BCE}} + L_{\text{IOU}} \tag{19}$$

where Loss is the total loss. L_{BCE} and L_{IOU} are the BCE loss and IOU loss, respectively. Technically, the BCE loss and IOU loss are formulated as follows:

$$L_{\text{BCE}} = -\sum_{x=1}^{H}\sum_{y=1}^{W}\left[G(x,y)\log\left(P(x,y)\right) + \left(1-G(x,y)\right)\log\left(1-P(x,y)\right)\right] \tag{20}$$

$$L_{\text{IOU}} = 1 - \frac{\sum_{x=1}^{H}\sum_{y=1}^{W} P(x,y)\,G(x,y)}{\sum_{x=1}^{H}\sum_{y=1}^{W}\left[P(x,y)+G(x,y)-P(x,y)\,G(x,y)\right]} \tag{21}$$

where W and H are the width and height of the images, respectively. $G(x, y)$ and $P(x, y)$ are the ground-truth label and the predicted saliency label at the location (x, y), respectively.

It is worth noting that we upsample the supervised features and use ground truth with different spatial resolutions to get the final predicted map and enhance LPENet's performance.

IV. Experiments

A. Experimental Setting and Implementation Details

We conducted comparative experiments on the SSP2000 dataset [49]. The SSP2000 dataset is the first internal surface defect dataset of cold-drawn pipes collected from a pipeline robot's perspective. It contains 2000 real-world images of internal surface defects. Following the experimental settings of Chen et al. [7], the dataset was divided into a training set of 1400 images and a testing set of 600 images.

To ensure that our method is more adaptable to the internal surface detection scenario, no preprocessing operations were applied to the 2000 images in the dataset. For a fair comparison, all experiments were implemented using the PyTorch framework under Ubuntu 20.04 with Python 3.8. The training was performed on a single NVIDIA GeForce RTX 4060Ti GPU, employing the classical stochastic gradient descent (SGD) optimizer. The initial learning rate of the network was set to 0.025, and the weight decay was set to 0.0001.

B. Evaluation Metric

To validate the effectiveness of the proposed LPENet, we adopt eight widely recognized evaluation metrics for SOD accuracy, which include S-measure (S_α), maximum and adaptive E-measure ($E_\xi^{\max}$ and E_ξ^{adp}), maximum, adaptive, mean, and weighted F-measure (F_β^{adp}, $F_\beta^{\max}$, F_β^{mean}, and F_β^{w}), and mean absolute error (MAE) to quantitatively assess the performance of the comparative experiments. For detection speed, we adopted frames per second (FPS), which denotes the average number of images tested per second. The higher the FPS, the faster the testing speed. In addition, the precision–recall (PR) curve and F-measure curve were also utilized for quantitative evaluation.

C. Comparison With State of the Art

We compared LPENet with 17 representative state-of-the-art SOD methods proposed in recent years, including six lightweight methods (CorrNet [40], FSMI [23], MobileSal [50], ADMNet [46], MINet [10], and SGGNet [30]), and 11 nonlightweight methods (EDRNet [3], DACNet [51], C2FNet [52], ICON [53], FEDER [54], LSRNet [55], HitNet [56], BBRF [57], SINet [58], SRPCNet [7], and RAMENet [21]). For all methods, the results were generated from publicly available source codes under their parameter settings.

1) Quantitative Results and Analysis: Table I presents the quantitative results of our method and 17 existing SOD approaches on the SSP2000 dataset. This table reports model size, parameter count, flops, eight accuracy metrics, and FPS. Notably, our method outperforms all other lightweight methods, achieving superior performance in both accuracy and inference speed. Compared with nonlightweight methods, our proposed LPENet also demonstrates competitive advantages. Although some accuracy metrics of LPENet are slightly lower than those of the best model, LPENet brings out the best effect among the nonlightweight methods; it achieves fast and accurate detection with significantly fewer parameters and a smaller model size. Specifically, compared with SRPCNet, the number of parameters and model size are reduced by 91.28% and 91.22%, respectively, making LPENet more suitable for practical deployment and industrial applications. Furthermore, the detection speed is improved by 566.81% over SRPCNet.

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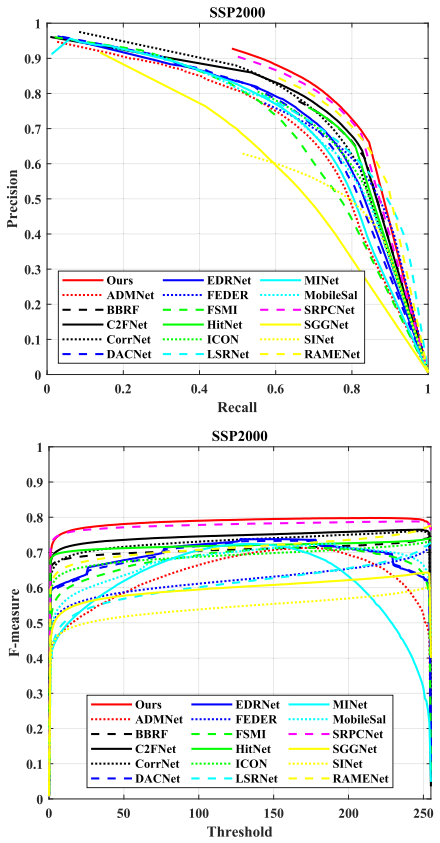


Fig. 8. PR curves and F-measure of different methods on the SSP2000 dataset.

In summary, as the first lightweight method for detecting internal surface defects in cold-drawn steel pipes, LPENet achieves an excellent balance between detection accuracy and inference speed. We also plot the PR curve and F-measure curve in Fig. 8.

2) Visualization Result Analysis: Visualization results in Fig. 9 show the visual comparison between our method and several representative approaches, demonstrating the adaptability of our method to defect images under various challenges of defect characteristics. The results highlighted by the red boxes indicate that LPENet can effectively detect low-contrast object defects; those highlighted by the green boxes show that LPENet can accurately detect extremely reflective object defects; and those highlighted by the blue boxes reveal that LPENet can successfully detect small-sized object defects.

3) Detection Efficiency Analysis: Real-time performance is one of the key factors in industrial detection. To quantitatively evaluate the detection efficiency of different methods, we conducted tests in a factory by integrating the PISD robot system with several industrial SOD approaches. Specifically, the PISD robot is capable of moving inside the pipe at a speed of 0.4 m/s, while its five CMOS sensors capture images at 10 frames/s; in other words, the robot can acquire 50 defect images of the internal surface/s. For a cold-drawn steel pipe of 8 m in length, the robot requires 20 s to pass the pipe, during which it collects approximately 1000 images. These images are, then, compressed and transmitted in real time to an industrial PC equipped with the deployed models. For fairness, the image transmission is ignored, and the image transfer time is negligible. Given that LPENet achieves a processing speed of 140 frames/s, the model takes approximately 8 s to process 1000 images. Therefore, the actual detection speed for internal surface detection of an 8-m pipe is estimated to be 1 m/s, and the total detection time is 28 s. Table II reports the quantitative results of detection efficiency, showing the model processing speed, total detection time, and robot detection speed. From these quantitative results, it can be observed that the FPS directly affects the robot's actual detection efficiency. Nevertheless, LPENet can achieve superior detection efficiency compared to other industrial detection methods, while maintaining a lightweight deployment size. This outstanding performance can be attributed to the effective training strategy and the task-specific network design.

D. Ablation Studies

In this section, to validate the key components of our method, including the PFE-based feature extraction backbone and the overall module architecture, we conducted extensive ablation studies on the SSP2000 dataset to investigate the contribution of each component.

1) Effectiveness of PFE Backbone: To evaluate the effect of different feature extraction backbones, we designed six variants of LPENet, as summarized in Table III. Specifically, we replaced the PFE-based backbone with two large CNN-based backbones (ResNet50 and VGG16) and four real-time or lightweight backbones (ResNet-12, ResNet-18, MobileNetV2, and EfficientNet). These six variants are referred to as ResNet50-Net, VGG16-Net, ResNet12-Net, ResNet18-Net, Mobile-Net, and Efficient-Net, respectively. According to the results in Table III, compared with large backbones such as VGG16, LPENet's model size count is only 8.7% of VGG16-Net. Across all eight SOD evaluation metrics and in terms of detection speed, LPENet significantly outperforms VGG16-Net, with a similar trend observed for ResNet50. These quantitative results indicate that, for detecting internal surface defects in cold-drawn steel pipes, the backbone of LPENet satisfies lightweight deployment requirements while providing real-time and accurate detection. To further validate the effectiveness of the LPENet real-time backbone, we compared LPENet with four lightweight backbones. LPENet demonstrates competitive advantages across all eight evaluation metrics, while maintaining fewer parameters and a smaller model size. In terms of FPS, LPENet achieves improvements of 303% and 359% over Mobile-Net and Efficient-Net, respectively, and achieves 368.2% and 229.4% improvements over ResNet12-Net and ResNet18-Net. Overall, these quantitative analyses fully demonstrate the effectiveness and efficiency of the PFE-based real-time backbone in achieving fast, accurate, and lightweight industrial defect detection.

2) Effectiveness of Different Modules: To further verify the effectiveness of different modules in LPENet, we also use the same eight SOD evaluation metrics and conduct a series of ablation experiments, as shown in Table IV. We conducted ablation studies on three core components individually (the second–fourth modules). Compared with

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TABLE I
QUANTITATIVE COMPARISONS WITH 15 STATE-OF-THE-ART METHODS ON THE SSP2000 DATASETS

Model	Type	Backbone	Model Size(M)↓	Params(M)↓	Flops(G)↓	FPS ↑	Evaluation Metrics							
							$S_\alpha \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{adp} \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{adp} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^w \uparrow$	MAE↓
CorrNet	L.W	VGG16	15.6	4.1	21.1	51.0	0.8091	0.9372	0.8983	0.7347	0.6251	0.7063	0.7010	0.0027
FSMI	L.W	FSMI-Based	13.6	3.6	11.8	18.2	0.7734	0.9099	0.8235	0.6766	0.5059	0.6386	0.6082	0.0034
MobileSal	L.W	MobileNetV2	10.2	6.5	**0.5**	57.3	0.7750	0.9138	0.7507	0.6850	0.4604	0.6371	0.6216	0.0029
ADMNet	L.W	ADM-Based	**3.9**	**0.8**	**1.0**	52.3	0.7859	0.9309	0.7255	0.6852	0.4224	0.6025	0.5818	0.0032
MINet	L.W	MI-Based	**4.3**	**0.3**	**0.4**	12.8	0.7829	0.9296	0.6904	0.6978	0.4003	0.5773	0.5596	0.0033
SGGNet	L.W	MobileNetV2	10.3	2.7	1.38	6.8	0.7187	0.8703	0.7710	0.5881	0.4473	0.5570	0.5440	0.0038
EDRNet	N.L.W	ResNet	150.0	39.3	32.4	33.9	0.8031	0.9345	0.8675	0.7170	0.5802	0.6649	0.6467	0.0027
DACNet	N.L.W	ResNet	375.5	98.9	142.9	21.8	0.7991	0.9277	0.8647	0.7070	0.5702	0.6582	0.6396	0.0030
C2FNet	N.L.W	ResNet	172.4	44.9	18.2	54.0	0.8201	0.9460	**0.9232**	0.7456	**0.6709**	**0.7245**	**0.7230**	0.0026
ICON	N.L.W	Swin-B	369.6	52.8	52.8	41.4	0.7898	0.9371	0.8683	0.7049	0.5872	0.6708	0.6766	0.0027
FEDER	N.L.W	ResNet	169.0	42.1	36.0	31.7	0.7705	0.9317	0.7415	0.6986	0.4717	0.6024	0.6166	0.0032
LSRNet	N.L.W	ResNet	194.0	28.7	20.8	62.2	0.7509	0.9423	0.6511	0.7021	0.3964	0.5969	0.6075	0.0034
HitNet	N.L.W	PVTv2	98.2	25.7	16.8	19.3	0.8124	0.9421	0.9227	0.7222	0.6701	0.6993	0.7103	0.0026
BBRF	N.L.W	Swin-T	283.8	74.4	46.4	**57.9**	0.8137	0.9451	0.9119	0.7284	0.6486	0.6915	0.7033	0.0027
SINet	N.L.W	ViT	186.9	44.4	25.9	41.3	0.7189	0.9265	0.6909	0.5820	0.4153	0.5274	0.5452	0.0038
SRPCNet	N.L.W	VGG16	69.3	18.1	50.0	21.1	**0.8259**	**0.9389**	**0.9309**	**0.7678**	**0.7115**	**0.7570**	**0.7446**	**0.0023**
RAMENet	N.L.W	MobileViT	23.7	5.2	8.7	**58.4**	**0.8219**	**0.9551**	0.8633	**0.7572**	0.6023	0.6984	0.7116	**0.0024**
LPENet(Ours)	L.W	PFE-based	**6.1**	**1.6**	23.2	**140.0**	**0.8216**	**0.9491**	**0.9297**	**0.7774**	**0.6988**	**0.7659**	**0.7491**	**0.0025**

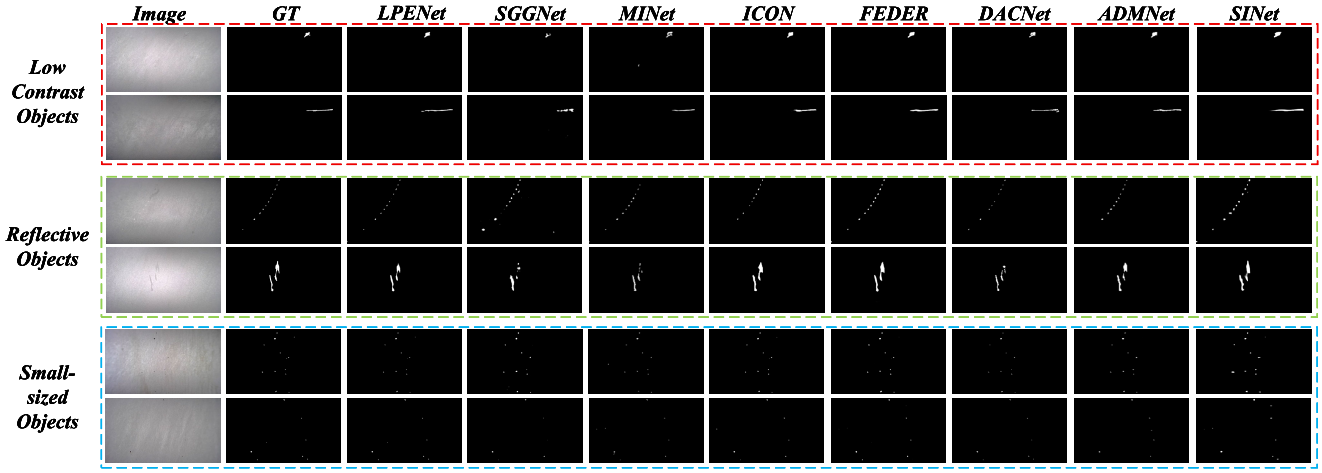


Fig. 9. Visual results on the SSP2000 dataset.

TABLE II
QUANTITATIVE RESULTS OF DETECTION EFFICIENCY

Metrics	LPENet(Ours)	SRPCNet	MINet	EDRNet	DACNet
Model Size(MB)	**6.1**	69.3	4.3	150.0	375.5
FPS	**140.0**	21.0	12.8	34.0	24.0
Total Detection time(s)	**28.0**	68.0	98.0	50.0	62.0
Detection Speed(m/s)	**1.0**	0.2	0.1	0.3	0.2

TABLE III
QUANTITATIVE COMPARISONS WITH BACKBONE VARIANTS

Variants	Model Size(MB)	Params(M)	FPS	$S_\alpha \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{adp} \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{adp} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^w \uparrow$	MAE↓
VGG16-Net	69.8	18.3	17.3	0.7861	0.8815	0.8825	0.7061	0.6587	0.6985	0.6724	0.0033
Mobile-Net	12.2	3.2	38.5	0.8068	0.9316	0.9128	0.7575	0.6705	0.7442	0.7251	0.0029
Efficient-Net	82.6	21.5	32.4	0.8127	0.9410	0.9005	0.7624	0.6559	0.7434	0.7297	0.0029
ResNet12-Net	44.12	11.53	29.9	0.8231	0.9347	0.9067	0.7670	0.6700	0.7452	0.7355	0.0023
ResNet18-Net	56.24	14.71	42.5	0.8034	0.9392	0.9270	0.7520	0.6803	0.7417	0.7187	0.0024
ResNet50-Net	259.2	64.6	31.1	0.8056	0.9346	0.9155	0.7586	0.6736	0.7445	0.7275	0.0030
LPENet(OURS)	**6.1**	**1.6**	**140.0**	**0.8216**	**0.9491**	**0.9297**	**0.7774**	**0.6988**	**0.7659**	**0.7491**	**0.0025**

the baseline results, each module consistently yields notable performance and improved detection accuracy, confirming their necessity and rationality in the overall design. This improvement is primarily attributed to the following factors: the MCE module effectively focuses on multiscale contextual information, enhancing the accurate perception of salient objects; the PGF module achieves cross-layer guidance through adaptive attention weights, enabling the fusion of shallow details and deep semantic information; and the BEA module aggregates contextual and boundary information through a multistep enhancement mechanism, enabling fine-grained detection. In addition, we further investigated the collaborative effects between different modules. Specifically, we tested various pairwise combinations among the fifth–seventh modules to evaluate their joint impact on detection performance. The results indicate that every

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TABLE IV
RESULTS OF THE ABLATION STUDIES

No.	Modules Combination			Evaluation metrics							
	MCE	PGF	BEA	$S_\alpha \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{adp} \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{adp} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{w} \uparrow$	MAE↓
1				0.7716	0.9132	0.8145	0.6813	0.5299	0.6578	0.6487	0.0034
2	✓			0.7939	0.9297	0.8429	0.7054	0.5697	0.6746	0.6764	0.0031
3		✓		0.7979	0.9370	0.9018	0.7219	0.6477	0.7106	0.7040	0.0027
4			✓	0.8070	0.9143	0.7841	0.7261	0.5277	0.6980	0.6755	0.0033
5	✓	✓		0.8008	0.9406	0.8767	0.7167	0.6138	0.6837	0.6910	0.0030
6	✓		✓	0.8167	0.9369	0.8795	0.7589	0.6221	0.7320	0.7204	0.0027
7		✓	✓	0.8129	0.9437	0.8618	0.7642	0.6039	0.7259	0.7199	0.0028
8	✓	✓	✓	**0.8216**	**0.9491**	**0.9297**	**0.7774**	**0.6988**	**0.7659**	**0.7491**	**0.0025**

TABLE V
RESULTS OF THE GENERALIZATION EXPERIMENTS

Methods	Dataset	$S_\alpha \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{adp} \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{adp} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{w} \uparrow$	MAE↓
EDRNet	NRSD-MN	0.7711	0.7821	0.7804	0.6897	0.6835	0.6814	0.6393	0.0708
BBRF	NRSD-MN	**0.7860**	**0.8256**	**0.8189**	**0.7492**	**0.7483**	**0.7438**	**0.6957**	0.0657
ADMNet	NRSD-MN	0.7687	0.7965	0.7952	0.6957	0.6932	0.6866	0.6429	0.0715
MINet	NRSD-MN	0.7807	0.8007	0.8016	0.7027	0.6983	0.6912	0.6505	0.0694
SRPCNet	NRSD-MN	0.7936	0.8145	0.8121	0.7202	0.7291	0.7123	0.6762	**0.0649**
LPENet(Ours)	NRSD-MN	**0.7941**	**0.8531**	**0.8456**	**0.7595**	**0.7537**	**0.7553**	**0.7068**	**0.0620**
EDRNet	SD-SALIENCY-900	0.8635	0.9562	0.9469	0.8544	0.8175	0.8328	0.8290	0.0279
BBRF	SD-SALIENCY-900	0.8533	0.9504	0.9501	0.8490	0.8329	0.8340	0.8285	0.0271
ADMNet	SD-SALIENCY-900	0.8624	0.9526	0.9425	0.8553	0.8191	0.8368	0.8240	0.0284
MINet	SD-SALIENCY-900	0.8428	0.9398	0.9323	0.8328	0.7876	0.8102	0.7922	0.0321
SRPCNet	SD-SALIENCY-900	0.8659	**0.9604**	**0.9545**	0.8664	0.8356	**0.8452**	**0.8383**	**0.0262**
LPENet(Ours)	SD-SALIENCY-900	**0.8679**	**0.9587**	**0.9556**	**0.8677**	**0.8442**	**0.8527**	**0.8405**	**0.0253**

two-module combination outperforms the individual modules, demonstrating their complementarity and cooperative benefits. Among them, the combination of MCE, PGF, and BEA modules achieves the best detection accuracy, which clearly highlights the advantages and potential of multimodule collaboration in addressing the detection task for internal surface defects of cold-drawn steel pipes.

E. Generalization Experiment

Most defect detection models are designed for specific industrial settings, which may lead to performance degradation when applied to different industrial scenarios. To further evaluate the generalization capability of LPENet in industrial applications, we conducted generalization experiments on the SD-SALIENCY-900 [59] and NRSD-MN [60] datasets. The SD-SALIENCY-900 dataset consists of 900 real steel strip surface defect images. Following the experimental setup of EDRNet, the dataset was reasonably divided into a training set of 540 images and a testing set of 360 images. For NRSD-MN, according to MCNet [60], 2086 images were used for training and 885 images for validation, while all natural steel rail surface images were employed as the testing set.

We compared LPENet with several outstanding industrial SOD methods (EDRNet, BBRF, ADMNet, MINet, and SRPCNet) that performed well on the SSP2000 dataset, as shown in Table I, and use the same eight evaluation metrics for quantitative assessment, as shown in Table V. The results demonstrate that LPENet achieves accuracy and efficiency both on the SD-SALIENCY-900 and NRSD-MN dataset effectively showing its generalization capability and robustness. These findings further confirm that LPENet can handle other types of industrial defect detection tasks effectively.

F. Limitation Analysis and Applicable Scope

Although the proposed detection method can achieve fast, accurate, and highly visualized detection with a lightweight model size, there is still room for improvement in terms of model deployment scalability. Specifically, the current detection pipeline requires deployment on an industrial PC with a GPU to process images captured by the PISD robot. However, the embedded board within the PISD robot also possesses certain computational capabilities. By deploying an optimized model on the NPU, efficient detection can be achieved directly on the robot. In future work, we plan to perform model pruning and optimization for embedded hardware deployment, reducing reliance on external computing platforms and enhancing the system's flexibility and applicability.

Additionally, through practical investigations at a cold-drawn steel pipe factory, we observed that large quantities of pipes are stacked in the inspection area, and reliance on manual inspection is time-consuming and subjective, highlighting the inefficiency of existing methods and validating the necessity of the proposed detection approach. While the current detection system can effectively handle stacked cold-drawn steel pipes, as the industry moves toward smart factory implementations, production increasingly emphasizes continuity across each manufacturing stage. Therefore, we aim to develop an automated online detection system for the production line, facilitating real-time detection and improving both the intelligence and overall efficiency of the manufacturing process.

V. CONCLUSION

This article presents a novel detection approach for real-time and accurate detection of internal surface defects in cold-drawn steel pipes. We design an efficient detection model, termed LPENet, which, integrated with the PISD robot and the TDS, achieves an optimal balance between detection efficiency and accuracy. Comprehensive experiments are conducted by comparing our method with several existing industrial-grade detection approaches using the PISD robot in real-world testing scenarios. The results demonstrate that our approach attains a detection speed of 1 m/s, thereby significantly advancing detection efficiency. Moreover, LPENet maintains a lightweight model size suitable for deployment, while effectively addressing challenges commonly encountered in cold-drawn steel pipe detection, such as low contrast, surface reflections, and small defect targets. Extensive evaluations on two industrial datasets highlight the superior performance of the proposed method. In future work, we aim to extend the deployment of the proposed method to embedded board-level systems and edge devices, and we anticipate that it will inspire further research in this field.

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